

CEN-204 Numerical Analysis

Textbook: Numerical Analysis

Mathematics of Scientific Computing

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-Numerical Analysis involves the study, development, and analysis of algorithms for obtaining numerical solutions to various mathematical problems.

Chapter-1 Mathematical Preliminaries

- **-Limit**

- If f is a real-valued function of a real variable, then the limit of the function f at c is defined as follows:
- $\lim_{x \rightarrow c} f(x) = L$
- means that f can be made to be as close to L as desired by making x sufficiently close to c .
- Or $|f(x) - L| < \varepsilon$ whenever $|x - c| < \delta$
- Or for each $\varepsilon > 0$ there is a $\delta > 0$ such that $0 < |x - c| < \delta$ makes $|f(x) - L| < \varepsilon$
- **Example:** Show that $\lim_{x \rightarrow 2} x^2 = 4$
- **Solution:** Let $\delta = -2 + \sqrt{4 + \varepsilon} > 0 \Rightarrow \delta(\delta + 4) = (-2 + \sqrt{4 + \varepsilon})(2 + \sqrt{4 + \varepsilon}) = \varepsilon$

For each $\varepsilon > 0$ there is $\delta = -2 + \sqrt{4 + \varepsilon}$ ($0 < |x - 2| < \delta$) that makes $|x^2 - 4| < \varepsilon$

Example: $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist (left and right limits are not the same)

• The function $f(x)$ is said to be continuous at c if $\lim_{x \rightarrow c} f(x) = f(c)$

Example: $f(x) = x^2$ is continuous at $x = 2$.

Example: $f(x) = \frac{|x|}{x}$ is not continuous at $x = 0$.

• Derivative of $f(x)$ at c is defined by the equation

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$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

If $f(x)$ is a function for which $f'(c)$ exists, we say $f(x)$ is differentiable at c . If $f(x)$ is differentiable at c , then $f(x)$ must be continuous at c . But the

reverse is not true.

Proof: $f(x)=|x|$ is a continuous function but not a differentiable function.

$$\lim_{x \rightarrow c} [f(x) - f(c)] = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} (x - c) = f'(c) \lim_{x \rightarrow c} (x - c) = f'(c) \cdot 0$$

$\Rightarrow$ if $f'(c)$ exists then $f(x)$ is continuous at c .

- The set of all functions that are continuous on the real line R is denoted by $C(R)$
- The set of functions for which $f'(x)$ is continuous on R is denoted by $C^1(R)$
- The set of functions for which $f''(x)$ is continuous on R is denoted by $C^2(R)$
- The set of functions for which $f^n(x)$ is continuous on R is denoted by $C^n(R)$
- $C^\infty(R)$ is the set of functions, each of whose derivatives is continuous.

$$\Rightarrow C^\infty(R) \subset \dots \subset C^n(R) \subset \dots \subset C^3(R) \subset C^2(R) \subset C^1(R) \subset C(R)$$

- $\Rightarrow C^n[a, b]$ To be the set of functions $f(x)$ for which $f^{(n)}(x)$ exists and is continuous on the interval $[a, b]$.

Taylor's Theorem

Taylor's Theorem with Lagrange Remainder

-If $f(x) \in C^n[a, b]$ and $f^n(x)$ exists on (a, b) , then for any points c and x in $[a, b]$,

$$f(x) = \sum_{k=0}^n \frac{1}{k!} f^k(c) (x-c)^k + E_n(x)$$

$$E_n(x) = \frac{1}{(n+1)!} f^{n+1}(\xi) (x-c)^{n+1} \text{ where } \xi \text{ is a point between } c \text{ and } x.$$

When $c=0$ $f(x) = \sum_{k=0}^n \frac{1}{k!} f^k(0) x^k + E_n(x)$ becomes Maclaurin series

$$\Rightarrow E_n(x) = \frac{1}{(n+1)!} f^{n+1}(\xi) x^{n+1}$$

Example: Find the Taylor series of $f(x)=\ln(x)$ for $a=1$, $b=2$, and $c=1$.

Solution:

$$f'(x)=x^{-1}, \quad f^{(2)}=-x^{-2}, \quad f^{(3)}=2x^{-3}, \quad f^{(4)}=-6x^{-4}, \quad f^{(5)}=24x^{-5}$$

$$\Rightarrow f^{(k)}(x)=(-1)^{k-1}(k-1)!x^{-k}$$

$$f(x)=\ln(x)=\sum_{k=0}^n \frac{1}{k!} f^{(k)}(c)(x-c)^k + E_n(x) = \sum_{k=0}^n \frac{(-1)^{k-1}}{k} (x-1)^k + E_n(x)$$

$$E_n(x) = \frac{(-1)^n}{(n+1)} \xi^{-(n+1)} (x-1)^{n+1} \quad (1 < \xi < x)$$

$$|E_n(x)| = \frac{1}{(n+1)} \xi^{-(n+1)} (x-1)^{n+1} < \frac{(x-1)^{n+1}}{(n+1)}$$

Example: Assume that we want to compute $\ln(2)$ with the formulae given in the previous example. We want accuracy to be less than 10^{-8} . How many terms do we need to use?

Solution: $|E_n(x)| < \frac{(x-1)^{n+1}}{(n+1)} = \frac{(2-1)^{n+1}}{n+1}$

$$\Rightarrow 10^{-8} < \frac{1}{n+1} \Rightarrow (n+1) > 10^8 \Rightarrow n \geq 10^8 = 100 \text{ million terms.}$$

Example: How many terms do we need to use to compute $\ln(1.5)$ with the same accuracy?

Solution:

$$\Rightarrow 10^{-8} < \frac{(1.5-1)^{n+1}}{n+1} = \frac{0.5^{n+1}}{n+1} \Rightarrow n \geq 22$$

Mean Value Theorem

If $f(x)$ in $C[a,b]$ and $f'(x)$ exist on (a,b) , then for x and c in $[a,b]$

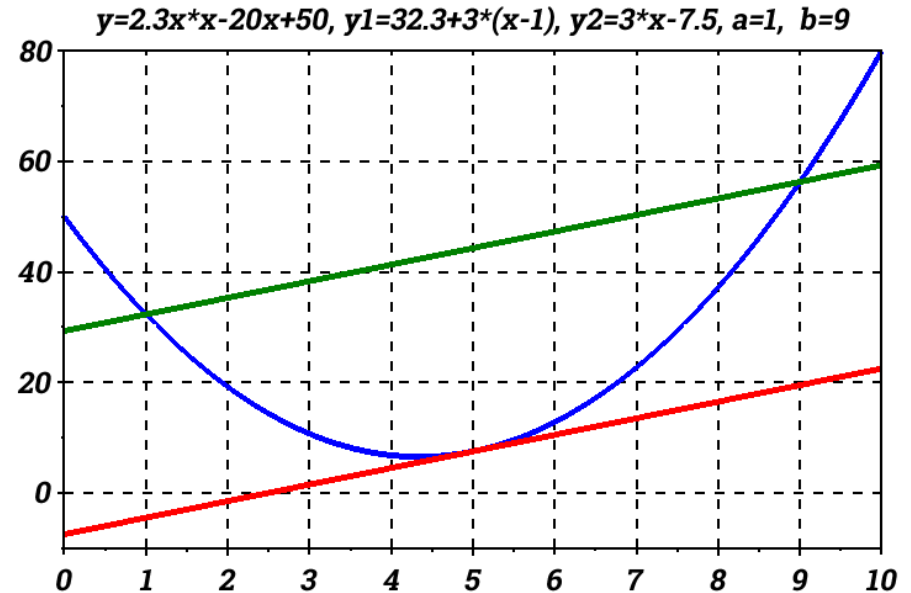
Zeroth order Taylor series expansion

$f(x) = f(c) + f'(\xi)(x-c)$ Where ξ is between c and x .

If $x=b$ and $c=a$ then

$f(b) - f(a) = f'(\xi)(b-a)$ where $a < \xi < b$

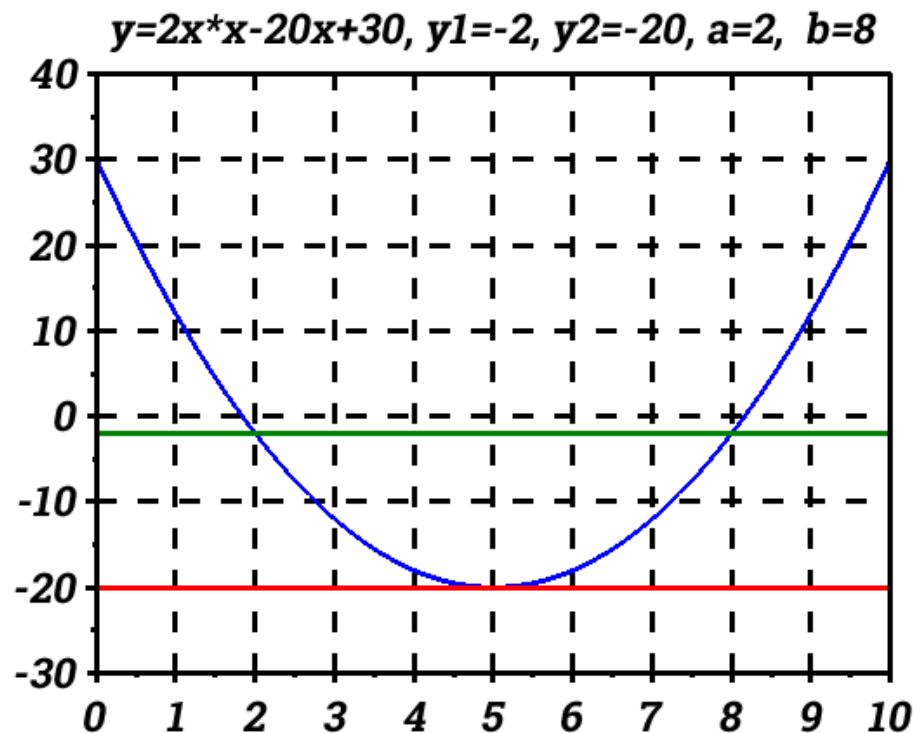
$$\Rightarrow f'(\xi) = \frac{f(b) - f(a)}{b - a}$$



$$f(a) = 32.3, \quad f(b) = 56.3, \quad (f(b) - f(a)) / (b - a) = 3$$

Rolle's Theorem

-If $f(x)$ in $C[a,b]$, if $f^{(1)}(x)$ exist on (a,b) , and if $f(a)=f(b)$, then $f'(\xi)=0$ for some ξ in (a,b) .



$$f(a)=f(b)=-2, \quad (f(b)-f(a))/(b-a)=0$$

Taylor's Theorem with Integral Remainder

If $f \in C^{n+1}[a, b]$ then for any points x and c in $[a, b]$,

$$f(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(c)(x-c)^k + R_n(x) \quad \text{where}$$

$$R_n(x) = \frac{1}{n!} \int_c^x f^{(n+1)}(t)(x-t)^n dt$$

Proof: let $u = \frac{(x-t)^n}{n!}$ and $dv = f^{(n+1)}(t) dt \Rightarrow v = f^{(n)}(t)$ and $du = \frac{-(x-t)^{n-1}}{(n-1)!} dt$

$$\Rightarrow R_n(x) = \frac{1}{n!} \int_c^x f^{(n+1)}(t)(x-t)^n dt = \int_c^x u dv = uv \Big|_c^x - \int_c^x v du$$

$$\Rightarrow R_n(x) = \frac{(x-t)^n}{n!} f^{(n)}(t) \Big|_c^x + \frac{1}{(n-1)!} \int_c^x f^{(n)}(t)(x-t)^{n-1} dt = \frac{-(x-c)^n}{n!} f^{(n)}(c) + R_{n-1}(x)$$

$$\Rightarrow R_n(x) = -\frac{(x-c)^n}{n!} f^n(c) - \frac{(x-c)^{n-1}}{(n-1)!} f^{n-1}(c) + R_{n-2}(x)$$

If we repeat integration, we get

$$\Rightarrow R_n(x) = -\sum_{k=1}^n \frac{f^k(c)}{k!} (x-c)^k + R_0(x)$$

$$\Rightarrow R_0(x) = \int_c^x f'(t) dt = f(t) \Big|_c^x = f(x) - f(c)$$

$$\Rightarrow f(x) = \sum_{k=0}^n \frac{f^k(c)}{k!} (x-c)^k + R_n(x)$$

Q.E.D.

Alternative Form of Taylor's Theorem

If $f(x) \in C^{n+1}[a, b]$, then for any points x and $(x+h)$ in $[a, b]$,

$$f(x+h) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(x) h^k + E_n(h)$$

$$E_n(h) = \frac{1}{(n+1)!} f^{(n+1)}(\xi) h^{n+1} \quad \text{where } \xi \text{ lies between } x \text{ and } (x+h).$$

$x+h \rightarrow x$ and $x \rightarrow c$

substitute $x+h$ for x and x for c

Taylor's Theorem in two Variables

If $f(x, y) \in C^{n+1}([a, b] \times [c, d])$, then for any points $(x+h)$ and $(y+k)$ in $[a, b] \times [c, d] \subseteq R^2$,

$$f(x+h, y+k) = \sum_{i=0}^n \frac{1}{i!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^i f(x, y) + E_n(h, k) \text{ where}$$

$$E_n(h, k) = \frac{1}{(n+1)!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^{n+1} f(x+\theta h, y+\theta k) + \text{ where } 0 < \theta < 1$$

The meaning of the terms: $\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^0 f(x, y) = f(x, y)$

$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^1 f(x, y) = \left(h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right)(x, y)$$

$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f(x, y) = \left(h^2 \frac{\partial^2 f}{\partial x^2} + 2hk \frac{\partial^2 f}{\partial x \partial y} + k^2 \frac{\partial^2 f}{\partial y^2} \right)(x, y) \text{ and so on.}$$

1.2 Orders of Convergence and Additional Basic Concepts

-In numerical calculations, it often happens that the answer to a problem is not produced all at once. Rather, a sequence of approximate answers is produced.

Convergent Sequences:

We write $\lim_{n \rightarrow \infty} x_n = L$ if there corresponds to each positive ε a real number r such that $|x_n - L| < \varepsilon$ whenever $n > r$. (Here n is an integer number.)

Example: Show that $\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$

Solution:

$$\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1 \quad \text{because} \quad \left| \frac{n+1}{n} - 1 \right| < \varepsilon \quad \text{whenever} \quad n > \varepsilon^{-1} = r$$

Example: $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$. If we compute the sequence $x_n = \left(1 + \frac{1}{n}\right)^n$

$$x_1 = 2.000000, \quad x_{10} = 2.593742, \quad x_{30} = 2.674319, \quad x_{50} = 2.691588, \\ x_{1000} = 2.716924, \quad e = 2.7182818$$

This is an example of a sequence that is converging slowly. Using double-precision computations, we find numerical evidence that

$$\frac{|x_{n+1}-e|}{|x_n-e|} \rightarrow 1 \quad \text{This property is worse than linear convergence.}$$

Example: $x_{n+1} = x_n - (x_n^2 - 2) \frac{x_n - x_{n-1}}{x_n^2 - x_{n-1}^2} \rightarrow \sqrt{2} \quad (\text{converge to } \sqrt{2})$

Let $x_1=2, \quad x_2=1.5, \quad \rightarrow x_3=1.428571, \quad x_4=1.414634,$

$x_5=1.414244, \quad x_6=1.414216, \quad \sqrt{2}=1.414213562$

Using double-precision computations, we find numerical evidence that

$$\frac{|x_{n+1}-\sqrt{2}|}{|x_n-\sqrt{2}|} \leq 0.77 \quad \text{which is called superlinear convergence.}$$

Example:

$$\left\{ \begin{array}{l} x_1 = 2, \\ x_{n+1} = \frac{1}{2}x_n + \frac{1}{x_n} \end{array} \quad n \geq 1 \right\} \quad \text{converges to } \sqrt{2}$$

$$\rightarrow x_2 = 1.5, \quad x_3 = 1.416667, \quad x_4 = 1.414216, \quad \sqrt{2} = 1.414213562$$

Using double-precision computations, we find numerical evidence that

$$\frac{|x_{n+1} - \sqrt{2}|}{|x_n - \sqrt{2}|^2} \leq 0.36 \quad \text{which is called quadratic convergence.}$$

Orders of Convergence

-Let $\lim_{n \rightarrow \infty} x_n = x^*$. We say that the rate of convergence is at least linear if there are constant $C < 1$ and an integer N such that

$$|x_{n+1} - x^*| \leq C |x_n - x^*| \quad (n \geq N)$$

-We say that the rate of convergence is at least superlinear if there exist a sequence ε_n tending to 0 and an integer N such that

$$|x_{n+1} - x^*| \leq \varepsilon_n |x_n - x^*| \quad (n \geq N)$$

-The rate of convergence is at least quadratic if there are constant C (not necessarily less than one) and an integer N such that

$$|x_{n+1} - x^*| \leq C |x_n - x^*|^2 \quad (n \geq N)$$

-In general, if there are positive constant C and α and an integer N such that

$$|x_{n+1} - x^*| \leq C |x_n - x^*|^\alpha \quad (n \geq N)$$

we say that the rate of convergence is of order α at least.

Big O and Little o Notation

Let x_n and α_n be two different sequences. We write $x_n = O(\alpha_n)$ if there are constants C and n_0 such that $|x_n| \leq C|\alpha_n|$ when $n \geq n_0$. Here we say that x_n is “Big oh” of α_n .

-The equation $x_n = o(\alpha_n)$ means that $\lim_{n \rightarrow \infty} (x_n / \alpha_n) = 0$. Here we say that x_n is “little oh” of α_n .

Example: $x_n = \frac{n+1}{n^2}$, $\alpha_n = \frac{1}{n} \Rightarrow x_n = O(\alpha_n)$

Example: $x_n = \frac{1}{n \ln(n)}$, $\alpha_n = \frac{1}{n} \Rightarrow x_n = o(\alpha_n)$

Example: $x_n = \frac{1}{n \ln(n)}$, $\alpha_n = \frac{1}{n} \Rightarrow x_n = o(\alpha_n)$

Example: $x_n = \frac{5}{n} + e^{-n}$, $\alpha_n = \frac{1}{n} \Rightarrow x_n = O(\alpha_n)$

Example: $x_n = e^{-n}$, $\alpha_n = \frac{1}{n^2}$ $\Rightarrow x_n = o(\alpha_n)$

Example: $x_n = \ln(2) - \sum_{k=1}^{n-1} (-1)^{k-1} \frac{1}{k}$, $\alpha_n = \frac{1}{n}$ $\Rightarrow x_n = O(\alpha_n)$

$$f(x) = \ln(x), \quad x=2, c=a=1, b=2 \Rightarrow x_n = E_{n-1}(x) = \frac{1}{n} (-1)^n \xi^{-n} (x-1)^n$$

$$\Rightarrow x_n = E_{n-1}(2) = \frac{1}{n} (-1)^n \xi^{-n} \quad 1 < \xi < 2$$

Example: $x_n = e^x - \sum_{k=0}^{n-1} \frac{1}{k!} x^k$, $\alpha_n = \frac{1}{n!}$, $|x| \leq 1 \Rightarrow x_n = O(\alpha_n)$

$$f(x) = e^x, \quad c=0, a=-1, b=1 \Rightarrow x_n = E_{n-1}(x) = \frac{1}{n!} e^{\xi} \Rightarrow x_n \leq \frac{e}{n!} \quad -1 \leq \xi \leq 1$$

-O and o notations can be used also for functions.

Example: $\sin(x) = x - \frac{x^3}{6} + O(x^5) \quad (x \rightarrow 0)$

Means that $|\sin(x) - x + \frac{x^3}{6}| \leq C|x^5|$ as $x \rightarrow 0$ C is a positive constant

-An equation of the form $f(x) = O(g(x)) \quad (x \rightarrow \infty)$ means that there exist constants r and C so that $|f(x)| \leq C|g(x)|$ whenever $x \geq r$

Example: $\sqrt{x^2+1} = O(x) \quad x \rightarrow \infty$ since $\sqrt{x^2+1} \leq 2x$ when $x \geq 1$

-In general, we write $f(x) = O(g(x)) \quad (x \rightarrow x^*)$ when there is a positive constant C and a neighborhood of x^* such that $|f(x)| \leq C|g(x)|$ in that neighborhood. Similarly, $f(x) = o(g(x)) \quad (x \rightarrow x^*)$ means that

$$\lim_{x \rightarrow x^*} [f(x)/g(x)] = 0$$

Mean Value Theorem for Integrals

Theorem: Let u and v be continuous real-valued functions on an interval $[a,b]$, and suppose that $v \geq 0$. Then there exists a point ξ in $[a,b]$ such that

$$\int_a^b u(x) v(x) dx = u(\xi) \int_a^b v(x) dx = I u(\xi)$$

Proof: Let α and β denote the least and greatest value of $u(x)$ on $[a,b]$, respectively. Then

$$\alpha \leq u(x) \leq \beta \quad (a \leq x \leq b) \quad \Rightarrow \quad \alpha \leq u(\xi) \leq \beta$$

since $v(x) \geq 0$ we have

$$\begin{aligned} \alpha v(x) &\leq u(x) v(x) \leq \beta v(x) \quad (a \leq x \leq b) & \text{Let } I &= \int_a^b v(x) dx \\ \Rightarrow \int_a^b \alpha v(x) dx &\leq \int_a^b u(x) v(x) dx \leq \int_a^b \beta v(x) dx \end{aligned}$$

$$\Rightarrow \alpha I \leq \int_a^b u(x) v(x) dx \leq \beta I. \quad \text{If } I = 0 \Rightarrow v(x) = 0. \quad \text{The result is trivial.}$$

$$\text{If } I \neq 0 \quad \Rightarrow \alpha \leq I^{-1} \int_a^b u(x) v(x) dx \leq \beta$$

By the intermediat-value theorem for continuous functions, there exists a point ξ in $[a,b]$ for which

$$u(\xi) = I^{-1} \int_a^b u(x) v(x) dx$$

Chapter-2 Computer Arithmetic

2.1 Floating-Point Numbers and Roundoff Errors

-Most computers deal with real numbers in the binary system.

$$(427.325)_{10} = 4 * 10^2 + 2 * 10^1 + 7 * 10^0 + 3 * 10^{-1} + 2 * 10^{-2} + 5 * 10^{-3}$$

$$(1001.11101)_2 = 1 * 2^3 + 0 * 2^2 + 0 * 2^1 + 1 * 2^0 + 1 * 2^{-1} + 1 * 2^{-2} + 1 * 2^{-3} + 0 * 2^{-4} + 1 * 2^{-5} \\ = (9.90625)_{10}$$

-The word length of the computer places restriction on the precision with which real numbers can be represented.

-Even 1/10 cannot be stored exactly in the computer.

$$1/10 = (0.0001\ 1001\ 1001\ 1001\ 1001\ 1001\ \dots)_2$$

If we print 1/10 out to 40 decimal places, we obtain the following result:

0.100000000149011611938476562500000000000000

-There are two conversions: From decimal to binary, and from binary to decimal. Because of conversions, there will be errors.

-The product of two numbers that have eight digits to the right of the decimal point will be a number that has 16 digits to the right of the decimal point. (need rounding)

-Rounding to the nearest number

Consider a positive decimal number x of the form $0.a_1a_2a_3\cdots a_{m-2}a_{m-1}a_m$ with m digits to the right of the decimal point. One rounds x to n decimal places ($n < m$) in a manner that depends on the value of $(n+1)$ st digit. If the digit is less than 5, the digits after the n th decimal place, are discarded. Otherwise, n th digit is increased by one and the remaining digits are discarded.

Example: Seven-digit numbers are rounded to four-digit numbers:

0.1735499 $\rightarrow$ 0.1735

0.9999500 $\rightarrow$ 1.0000

0.4321609 $\rightarrow$ 0.4322

If x is the number and $\tilde{x}$ is the rounded number, then

$|x - \tilde{x}| \leq \frac{1}{2} 10^{-n}$ where n is the number of decimal places after the decimal point for the rounded number.

Truncation: Discard all the digits beyond the n th digit.

If x : number, $\hat{x}$: truncated number $\Rightarrow |x - \hat{x}| < 10^{-n}$

Normalized Scientific Notation

$$732.5051 = 0.7325051 * 10^3$$

$$-0.005612 = -0.5612 * 10^{-2}$$

In general, a nonzero real number x can be represented in the form:

$$x = \pm r 10^n, \quad \frac{1}{10} \leq r < 1, \quad n \text{ is an integer number}$$

For binary numbers:

$$x = \pm q 2^m, \quad \frac{1}{2} \leq q < 1, \quad q \text{ is called mantissa and the integer } m \text{ is the exponent.}$$

A slightly different scientific notation:

$$q = (1.f)_2 \Rightarrow 1 \leq q < 2$$

Hypothetical Computer Marc-32 (Same as float number in the programming languages such as C/C++)

Sign of mantissa 1 bit (s)	Biased exponent (e) 8 bits (unsigned integer)	Normalized Mantissa (f) 23 bits
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implicit radix point

$$q = (1.f)_2 \Rightarrow 1 \leq q < 2$$

$$m = e - 127$$

$$x = (-1)^s q 2^m \quad \text{Normalized floating point form.}$$

-Most real numbers are not precisely re-presentable within the Marc-32.

$$0 < e < (1111\ 1111)_2 = 2^8 - 1 = 255$$

$e=0$ and $e=255$ are reserved for special cases such as ± 0 , $\pm \infty$ and NaN

$$m = e - 127 \quad \Rightarrow -126 \leq m \leq 127$$

-Marc-32 can handle numbers as small as $2^{-126} \approx 1.2 * 10^{-38}$ and as large as $(2 - 2^{-23}) * 2^{127} \approx 3.4 * 10^{38}$

-Mantissa has 23 bits. Therefore, our machine numbers have a limited precision of roughly six decimal places, since the least significant bit in the mantissa represents unit of 2^{-23} (or approximately $1.2 * 10^{-7}$). Thus, numbers expressed with more than six decimal digits will be approximated when given as input to the computer.

$$\text{Zero:} \quad +0 = [0000\ 0000]_{16} \quad -0 = [8000\ 0000]_{16}$$

$$\text{Infinity:} \quad +\infty = [7F\ 80\ 0000]_{16} \quad -\infty = [FF\ 80\ 0000]_{16}$$

$$\text{Not a Number (NaN):} \quad 0/0, \quad \infty - \infty, \quad x + NaN \quad \Rightarrow e = 255, \quad f \neq 0$$

Machine Rounding

- In addition to rounding input data, rounding is needed after most arithmetic operations.
- The default rounding mode is round to nearest. The maximum error is half a unit of rounding in the least significant place.

Nearby Machine Numbers

Let $x = q * 2^m$, $1 \leq q < 2$, $-126 \leq m \leq 127$. What is the machine number closest to x ?

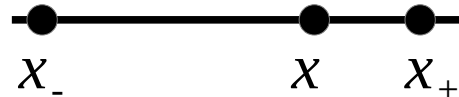
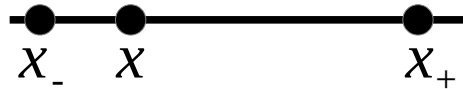
$x = (1.a_1 a_2 a_3 a_4 a_5 \dots a_{23} a_{24} a_{25} \dots)_2 * 2^m$ in which a_i is either 0 or 1.

- One nearby machine number is obtained by simply discarding the excess bits $a_{24} a_{25} a_{26}$. (called chopping)

$$\Rightarrow x_- = (1.a_1 a_2 a_3 a_4 a_5 \dots a_{21} a_{22} a_{23})_2 * 2^m$$

- Another nearby machine number lies to the right of x (rounding up)

$$\Rightarrow x_+ = ((1.a_1 a_2 a_3 a_4 a_5 \dots a_{21} a_{22} a_{23})_2 + 2^{-23}) * 2^m$$



$$|x - x_-| \leq \frac{1}{2} |x_+ - x_-| = \frac{1}{2} * 2^{m-23} = 2^{m-24}$$

The relative error is bounded as follows:

$$\left| \frac{x - x_-}{x} \right| \leq \frac{2^{m-24}}{q 2^m} = \frac{1}{q} 2^{-24} \leq 2^{-24}$$

-When x is closer to x_+ than to x_-

$$|x - x_+| \leq \frac{1}{2} |x_+ - x_-| = \frac{1}{2} * 2^{m-23} = 2^{m-24} \quad \Rightarrow \left| \frac{x - x_+}{x} \right| \leq \frac{2^{m-24}}{q 2^m} = \frac{1}{q} 2^{-24} \leq 2^{-24}$$

The relative error cannot be greater than 2^{-24}

If x is a nonzero real number and x^* is the machine number (marc-32) closest to x , then

$$\left| \frac{x - x^*}{x} \right| \leq 2^{-24}, \quad \text{let } \delta = \left(\frac{x^* - x}{x} \right) \quad \text{and } fl(x) = x(1 + \delta) \quad \Rightarrow |\delta| \leq 2^{-24}$$

-The notation $fl(x)$ is used to denote the floating point machine number x^* closest to x .

The number 2^{-24} is called the unit roundoff error for the marc-32.

Relative Error Analysis

Theorem: Let $x_0 x_1 x_2 x_3 \dots x_n$ be positive machine numbers in a computer whose unit roundoff error is ε . Then the relative roundoff error in computing $\sum_{i=0}^n x_i$ is at most $(1 + \varepsilon)^n - 1 \simeq n \varepsilon$

2.2 Absolute and Relative Errors: Loss of Significance

The absolute error $= |x - x^*|$ where x^* is approximation of x

The relative error $= \left| \frac{x - x^*}{x} \right|$

Loss of Significance

-Large relative error can occur after subtraction of two numbers that are close to each other.

Example: Let $x=0.3721478693$ and $y=0.3720230572$ (ten digit after the decimal point)

Do not represent additional accuracy

$x - y = 0.1248121000 * 10^{-3}$ (7 digits after the decimal point. 3 digits are lost.

If this calculation were to be performed in a decimal computer having a five-digit mantissa :

$$fl(x) = 0.37215$$

$$fl(y) = 0.37202$$

Three digits are lost.

$$fl(x) - fl(y) = 0.00013 = 0.13000 * 10^{-3}$$

$$\text{The relative error} = \left| \frac{x - y - [fl(x) - fl(y)]}{x - y} \right| = \left| \frac{0.0001248121 - 0.00013}{0.0001248121} \right| \simeq 4 \%$$

Subtraction of Nearly Equal Quantities

Example: $y = \sqrt{x^2 + 1} - 1$. Assume x is close to zero. It involves loss of significance for small values of x . How can we avoid this trouble?

Solution:

$$y = \sqrt{x^2 + 1} - 1 = \left(\sqrt{x^2 + 1} - 1 \right) \frac{\sqrt{x^2 + 1} + 1}{\sqrt{x^2 + 1} + 1} = \frac{x^2}{\sqrt{x^2 + 1} + 1}$$

Loss of Precision

Theorem: If x and y are positive normalized floating-point binary machine numbers such that $x > y$ and $2^{-q} \leq 1 - \frac{y}{x} \leq 2^{-p}$ Then at most q and at least p

significant binary bits are lost in the subtraction $x - y$.

Example: $y = x - \sin x$

Since $\sin x \simeq x$ for small values of x , this calculation will involve a loss of significance. How can this be avoided?

Solution: The Taylor series of $\sin x$ can be used.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} \dots$$

$$\Rightarrow y = x - \sin x = \frac{x^3}{3!} - \frac{x^5}{5!} + \frac{x^7}{7!} - \frac{x^9}{9!} \dots$$

We need to use only a couple of terms.

2.3 Stable and Unstable Computations: Conditioning

-A numerical process is unstable if small errors made at one stage of the process are magnified in subsequent stages and seriously degrade the accuracy of the overall calculation.

Example: $x_0 = 1, \quad x_1 = \frac{1}{3}, \quad x_n = \frac{13}{3}x_{n-1} - \frac{4}{3}x_{n-2} \quad n \geq 1$

In fact $x_n = \frac{1}{3^n}$

Proof (by induction): It is true for $n=0$ and $n=1$.

Assume it is true for $n \leq m$. Lets show that it is true for $n=m+1$.

$$\text{If } n=m \Rightarrow x_m = \frac{13}{3} x_{m-1} - \frac{4}{3} x_{m-2} = \frac{1}{3^m}$$

$$\text{If } n=m+1 \Rightarrow x_{m+1} = \frac{13}{3} x_m - \frac{4}{3} x_{m-1} = \frac{13}{3} \frac{1}{3^m} - \frac{4}{3} \frac{1}{3^{m-1}} = \frac{1}{3^{m-1}} \left[\frac{13}{9} - \frac{4}{3} \right] = \frac{1}{3^{m+1}}$$

Lets compute $x_n = \frac{13}{3} x_{n-1} - \frac{4}{3} x_{n-2}$ using marc-32 computer (float)

$$x_0 = 1.0000000$$

$$x_1 = 0.3333333 \quad (\text{seven correctly rounded significant digit})$$

$$x_2 = 0.1111112 \quad (\text{six correctly rounded significant digit})$$

$$x_3 = 0.00370375 \quad (\text{five correctly rounded significant digit})$$

$\vdots$

$$x_7 = 0.0005131 \quad (\text{one correctly rounded significant digit})$$

$$x_{14} = 0.9143735$$

$$x_{15} = 3.657493 \quad \text{incorrect with relative error } 10^8$$

This algorithm is unstable. Any error present in x_n is multiplied by $13/3$ in computing x_{n+1} . Error in x_1 may propagate in x_{15} with a factor $(13/3)^{14}$

-The general solution of the previous equation is

$$x_n = A \left(\frac{1}{3} \right)^n + B 4^n$$

Example:

If $x_0=1$, $x_1=4$ then $x_n=4^n$ (correct solution)

If we compute x_n using marc-32

$$x_1=4.000006, \quad x_{10}=1.048576 * 10^6, \quad x_{20}=1.099512 * 10^{12}$$

-The absolute errors are undoubtedly large as before but they are relatively negligible.

Example: (Numeric instability) Compute y_n

$$y_n = \int_0^1 x^n e^x dx \quad n \geq 0 \quad \text{Apply integration by part}$$

$$y_0 = \int_0^1 e^x dx = e - 1$$

$$y_{n+1} = \int_0^1 x^{n+1} e^x dx \quad \text{Let } u(x) = x^{n+1}, \quad e^x dx = dv \quad \Rightarrow v = e^x, \quad du = (n+1)x^n dx.$$

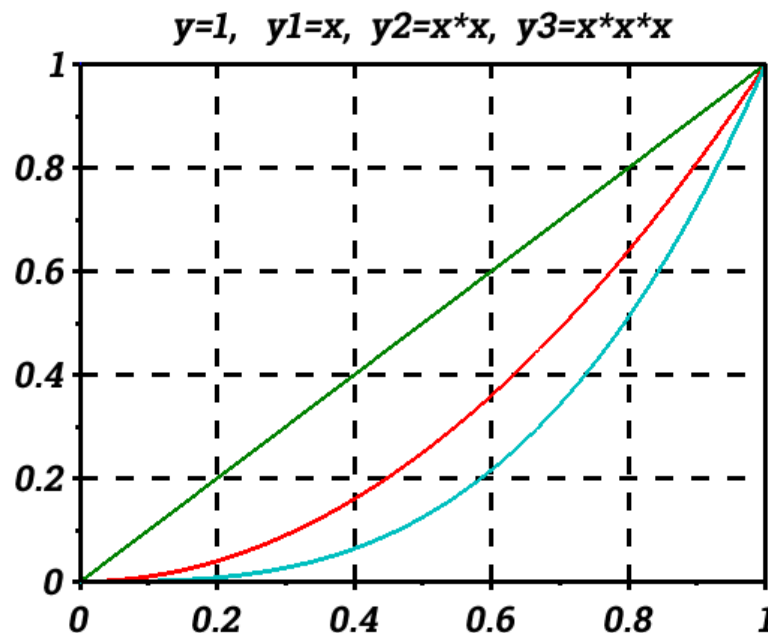
$$\Rightarrow y_{n+1} = x^{n+1} e^x \Big|_0^1 - (n+1) \int_0^1 x^n e^x dx = e - (n+1) y_n$$

$$\Rightarrow y_1 = 1, \quad y_2 = 0.7182817, \quad y_{11} = 1.422453,$$

$$y_{15} = 39711.43$$

in fact $y_1 > y_2 > y_3 > \dots > 0$

The error at nth step is multiplied by $(n+1)$ in computing y_{n+1} .



Conditioning

- A problem is ill conditioned if small changes in the data can produce large changes in the answer.
- For certain types of problems, a condition number can be defined.
- Condition Number for $f(x)$.**
- If x is perturbed slightly, what is the effect on $f(x)$?

$$\underbrace{f(x+h) - f(x)}_{\text{Error}} = hf'(\xi) \simeq \underbrace{hf'(x)}_{\substack{\text{Change in } x. \\ \text{Condition number for this problem}}} \quad (\text{mean value theorem})$$

Sometimes, the relative error is important.

$$\underbrace{\frac{f(x+h) - f(x)}{f(x)}}_{\text{Relative change in } f(x)} \simeq \frac{hf'(x)}{f(x)} = \underbrace{\left[\frac{xf'(x)}{f(x)} \right]}_{\text{Condition number for this problem}} \left(\frac{h}{x} \right) \rightarrow \text{Relative change in } x.$$

Example: Let $f(x)=\arcsin(x)$. Find a condition number for $f(x)$. (Relative change in $f(x)$.)

Solution:

$$\text{Condition Number} = \frac{xf'(x)}{f(x)} = \frac{x}{\sqrt{1-x^2} \arcsin x}$$

For x near 1 $\arcsin x \simeq \frac{\pi}{2}$ condition number $\rightarrow \infty$ Hence, small relative error in x may lead to large relative errors in $\arcsin(x)$ near $x=1$.

Chapter-3: Solution of Nonlinear Equations

-We want to find x such that $f(x)=0$.

Example: $f(x)=x-\tan(x)=0$

Example: $x-a*\sin(x)=b$

-There may be many approximate solutions even though the exact solution is unique. (Because of roundoff errors.)

Example: $P_4(x)=x^4-4x^3+6x^2-4x+1=(x-1)^4$

If you use `double`, you will find many zeros in the interval $[0.975, 1.035]$

3-1 Bisection (Interval Halving) Method

-If $f(x)$ is a continuous function on the interval $[a,b]$ and if $f(a)f(b)<0$, then $f(x)$ must have a zero in (a,b) .

Bisection Method:

- 1- Compute $c = 0.5 \cdot (a + b)$
- 2- If $f(a)f(c) < 0$ then $f(x)$ has a zero in $[a, c] \Rightarrow b \leftarrow c$ (assign c to b)
- 3- Else $\Rightarrow a \leftarrow c$ (assign c to a $f(x)$ has a zero in $[c, b]$)
- 4- Stop if $f(c) = 0$. (c is the zero of $f(x)$)
- 5- Go to step 1.

-It is quite unlikely that $f(c)$ will be exactly 0 in the computer because of roundoff errors.

Stopping Criteria (stop when one of them is satisfied)

- 1-The maximum number of steps (M)
- 2- $|f(c)| < \epsilon$ ϵ is a positive number.
- 3- $|b - c| < \delta$ δ is a positive number.